Cal Poly Pomona

FALL 2026 POINTS: 100

Computer Science 4800

Software Engineering
Project #5
UAV Perception System

Project Goal

Develop a ROS 2 perception component that detects objects of interest in a simulated UAV camera feed and maintains their identities across successive observations.

Scenario

A PX4-controlled UAV observes an environment using a simulated onboard camera. The system must identify objects belonging to specified classes and track individual objects as the UAV and objects move. A predefined flight trajectory or prerecorded sensor data may be provided so that the project can be developed independently of the path-planning teams.

Core Requirements

- Accept simulated camera images through ROS 2.
- Obtain UAV pose and state information from PX4 through MAVROS.
- Detect objects belonging to specified target classes.
- Produce a bounding box, class label, and confidence score for each detection.
- Assign persistent identifiers to detected objects.
- Associate detections with existing tracks across successive frames.
- Handle objects entering and leaving the camera's field of view.
- Handle temporary loss or occlusion of tracked objects.
- Remove stale tracks after a configurable period without observations.
- Publish detections and tracked objects through documented ROS 2 interfaces.
- Provide visualization of bounding boxes, object classes, confidence scores, and tracking IDs.
- Log detection and tracking results for subsequent analysis.
- Measure performance using metrics such as detection accuracy, track continuity, processing latency, and frame rate.
- Provide automated tests using repeatable prerecorded or simulated scenarios.

Teams may use an existing pretrained object detector, such as YOLO, rather than developing or training a neural network. The emphasis should be on the software architecture, integration, tracking, testing, and evaluation of the perception pipeline rather than on machine-learning model development.

Agile Development

Begin with a walking skeleton that receives camera images from the PX4 simulation through ROS 2 and publishes or displays the unmodified images. Subsequent sprints can integrate a pretrained object detector, add basic detection association and persistent tracking IDs, implement track lifecycle and occlusion handling, and improve performance and robustness. The product backlog should contain user stories such as "As a UAV operator, I want detected objects to retain consistent tracking IDs across video frames so that I can distinguish and monitor individual objects over time."

Potential Acceptance Criteria

Given a simulated environment or prerecorded test sequence, the system successfully detects specified object classes and maintains consistent identities for tracked objects within defined performance thresholds. Detection and tracking outputs conform to the specified ROS 2 interfaces; UAV state information is successfully obtained from PX4 through MAVROS; temporary object occlusions do not immediately terminate valid tracks; and the system continues operating safely when camera frames or vehicle-state messages are missing or delayed.